

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-2 LAB EXERCISE

Step 1 — Create Database

```
[mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.00 sec)

[mysql> USE bookflow_db;
Database changed
```

Step 2 — Create Books Table (DDL + Constraints)

```
mysql> CREATE TABLE Books (
->     book_id INT PRIMARY KEY,
->     title VARCHAR(100) NOT NULL,
->     isbn VARCHAR(20) UNIQUE,
->     published_year INT CHECK (published_year < 2027)
-> );
Query OK, 0 rows affected (0.01 sec)
```

Step 3 & 4 — Insert Books and Display

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

```
3 rows in set (0.00 sec)
```

Step 5, 6 & 7 — Members Table: Create, Insert, Display

```
mysql> CREATE TABLE Members (
->     member_id INT PRIMARY KEY,
->     full_name VARCHAR(100),
->     email VARCHAR(100) UNIQUE
-> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Members;
Empty set (0.00 sec)
```

```
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
3 rows in set (0.00 sec)
```

Step 8 — Create Loans Table (FOREIGN KEYS)

```
mysql> CREATE TABLE Loans (  
->     loan_id INT PRIMARY KEY,  
->     member_id INT,  
->     book_id INT,  
->     loan_date DATE,  
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

Step 9 & 10 — Insert 10 Loans and Display

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
-> (1, 101, 1, '2025-01-05'),  
-> (2, 102, 2, '2025-01-08'),  
-> (3, 103, 3, '2025-01-10'),  
-> (4, 101, 2, '2025-02-01'),  
-> (5, 102, 1, '2025-02-05'),  
-> (6, 103, 2, '2025-02-12'),  
-> (7, 101, 3, '2025-03-01'),  
-> (8, 102, 3, '2025-03-07'),  
-> (9, 103, 1, '2025-03-15'),  
-> (10, 101, 1, '2025-04-01');  
Query OK, 10 rows affected (0.00 sec)  
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)
```

Step 11 — JOIN Query (Who Borrowed What)

```
mysql> SELECT m.full_name AS Member_Name,  
->     b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m  
->     ON l.member_id = m.member_id  
-> INNER JOIN Books b  
->     ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

```
10 rows in set (0.00 sec)
```

Step 12 — GROUP BY Query (Books per Year)

```
mysql> SELECT published_year,
->      COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
+-----+-----+
| published_year | Total_Books |
+-----+-----+
|          1925 |           1 |
|          1949 |           1 |
|          1960 |           1 |
+-----+-----+
3 rows in set (0.00 sec)
```

Step 13–16 — Donation via TRANSACTION (Atomicity)

```
mysql> CREATE TABLE Donation_History (
->      donation_id INT PRIMARY KEY,
->      book_id INT,
->      donor_name VARCHAR(100),
->      donation_date DATE,
->      FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History
-> (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title           | isbn          | published_year |
+-----+-----+-----+-----+
|        1 | The Great Gatsby | 9780743273565 |          1925 |
|        2 | To Kill a Mockingbird | 9780061120084 |          1960 |
|        3 | 1984            | 9780451524935 |          1949 |
|        4 | Animal Farm      | 9780451526342 |          1945 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
|            1 |        4 | Raj Kumar  | 2026-07-29    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.01 sec)
```

Step 17 & 18 — Index on ISBN + Fast Search

```
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books
-> WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|        3 | 1984  | 9780451524935 |          1949 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Final State of All Tables

```
mysql> SHOW TABLES;
```

Tables_in_bookflow_db
Books
Donation_History
Loans
Members

```
4 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-07-29

```
1 row in set (0.00 sec)
```